DAVINE TECHNOLOGIES

TalentFlow AI

Technical Implementation & Architecture Report

AI-Powered Recruitment & Talent Management Platform

AWS Cloud Implementation & Architecture

Organization	Davine Technologies
Domain	AWS Cloud
Project	TalentFlow AI
AWS Region	Europe (Stockholm) eu-north-1
Infrastructure as Code	Terraform (AWS provider v5.100.0)
Document	Final Submission Report, Version 1.2
Date	30 October 2026

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1. Executive Summary & Scope

TalentFlow AI is an AI-powered recruitment and talent management platform that supports the complete recruitment lifecycle: candidate application, resume submission, AI-based screening, evaluation, shortlisting, interview scheduling, selection and hiring.

This project builds the AWS cloud foundation for the platform and specifies the full production architecture around it. The work is delivered as a **hybrid implementation**:

- **Implemented and verified:** the foundational VPC network, the resume storage bucket, the event-driven ingestion pipeline (EventBridge rule, SQS queue, Lambda trigger), recruiter alerting (SNS topic and subscription, CloudWatch alarm) and the Terraform configuration that codifies the base infrastructure.
- **Specified, not deployed:** the compute tier (EC2, Application Load Balancer, Auto Scaling), the managed database (Amazon RDS PostgreSQL) and the edge and API layer (Route 53, CloudFront, API Gateway). These are documented as production architecture specifications and were intentionally not left running during this phase, to avoid ongoing cloud charges.

Status labels used in this report

Label	Meaning
PASS (Console Verified)	The component exists in the AWS Management Console and is shown in a screenshot in this report
PASS (Terraform Validated)	Verified by Terraform initialisation and validation output shown in this report
Design Only	Architectural specification or future production scope; not deployed and not validated by code

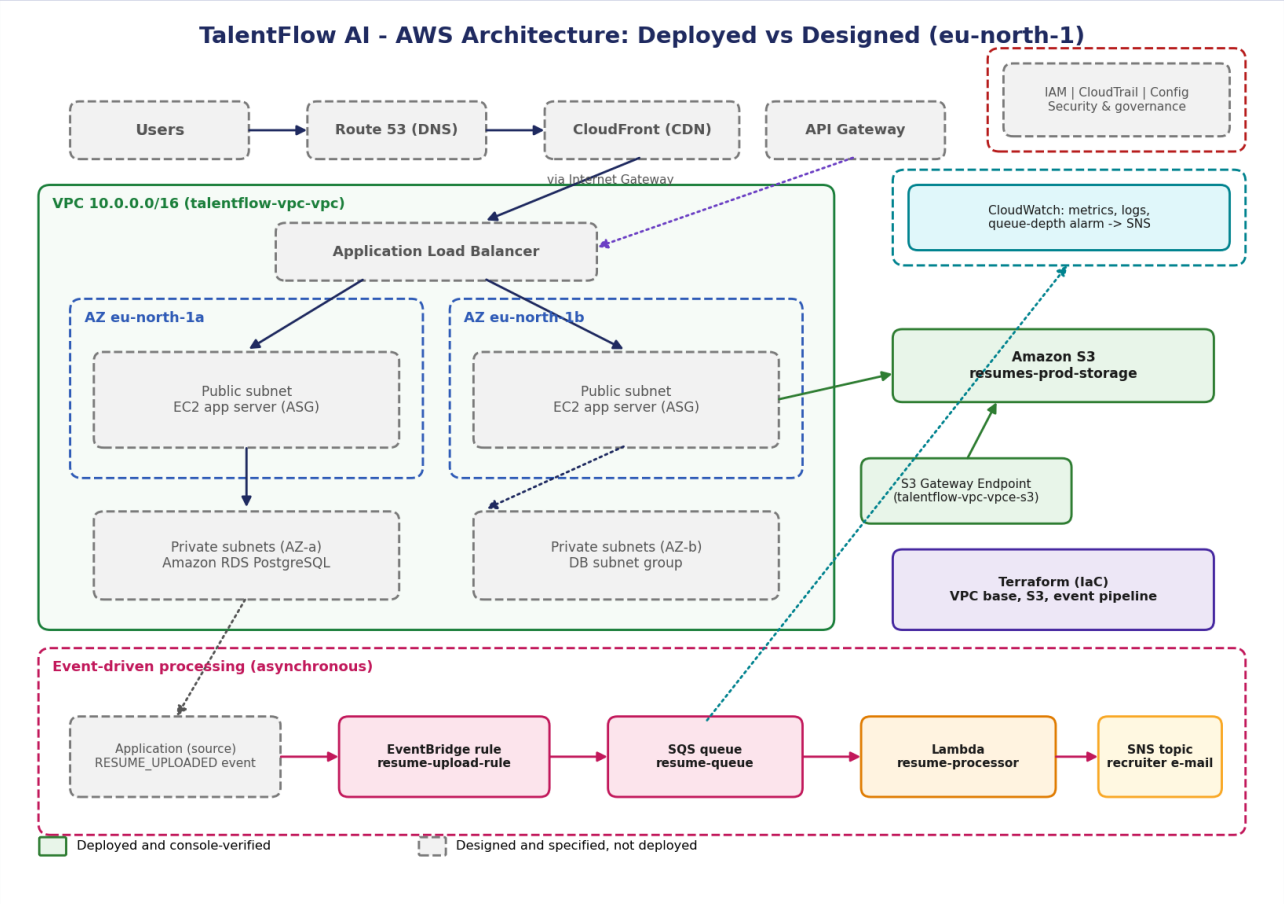
Implementation summary

Layer	Components	Status
Network	VPC 10.0.0.0/16, public and private subnets, Internet Gateway, S3 gateway endpoint	**PASS** (Console Verified)
Storage	S3 bucket `talentflow-resumes-prod-storage` with an uploaded resume	**PASS** (Console Verified)
Event pipeline	EventBridge rule, SQS queue, Lambda function with SQS trigger	**PASS** (Console Verified)
Alerting	SNS topic and e-mail subscription, CloudWatch queue-depth alarm	**PASS** (Console Verified)
Infrastructure as Code	Terraform init and validate	**PASS** (Terraform Validated)
Compute	EC2, Application Load Balancer, Auto Scaling Group	**Design Only**
Database	Amazon RDS PostgreSQL	**Design Only**
Edge and API	Route 53, CloudFront, API Gateway	**Design Only**
Governance	IAM review, CloudTrail, AWS Config, Budgets, Cost Explorer	**Design Only**
Containers	Docker, ECR, ECS/EKS	**Design Only** (roadmap)

2. System Architecture & Service Mapping

Target request path: Users, Route 53, CloudFront, Application Load Balancer, EC2 Auto Scaling Group, Amazon RDS.
Implemented event path: Application event, EventBridge, SQS, Lambda, SNS recruiter notification.
Control path: Terraform codifies the base infrastructure; CloudWatch monitors the queue.

Figure 1: Architecture, deployed and designed components



Solid components are deployed and console-verified. Dashed grey components are production architecture specifications that are not deployed.

AWS Service	Resource	Functional Role	Status
Amazon VPC	`talentflow-vpc-vpc` (10.0.0.0/16)	Network isolation boundary	**PASS** (Console Verified)
Internet Gateway	`talentflow-vpc-igw`	Public ingress and egress for public subnets	**PASS** (Console Verified)
S3 Gateway Endpoint	`talentflow-vpc-vpce-s3`	Private path from the VPC to S3	**PASS** (Console Verified)
Amazon S3	`talentflow-resumes-prod-storage`	Resume and document storage	**PASS** (Console Verified)
Amazon EventBridge	`talentflow-resume-upload-rule`	Routes RESUME_UPLOADED business events	**PASS** (Console Verified)
Amazon SQS	`talentflow-resume-queue`	Buffers resume-processing work	**PASS** (Console Verified)
AWS Lambda	`talentflow-resume-processor`	Resume worker, triggered by the SQS queue	**PASS** (Console Verified): function and trigger
Amazon SNS	`talentflow-recruiter-notifications`	Recruiter e-mail notifications	**PASS** (Console Verified)
Amazon CloudWatch	`talentflow-resume-queue-alarm`	Queue-depth alarm	**PASS** (Console Verified)
Terraform	AWS provider v5.100.0	Infrastructure as Code	**PASS** (Terraform Validated)
Amazon EC2	t3.micro, Amazon Linux 2023	Application servers	**Design Only**
Application Load Balancer	Public subnets, two AZs	Traffic distribution and health checks	**Design Only**
Auto Scaling Group	Launch template, 2 / 2 / 4	Elastic capacity	**Design Only**

AWS Service	Resource	Functional Role	Status
Amazon RDS (PostgreSQL)	db.t3.micro, private subnets	Managed relational database	**Design Only**
Amazon Route 53	Hosted zone	Domain-based access	**Design Only**
Amazon CloudFront	Distribution	Edge delivery	**Design Only**
Amazon API Gateway	Serverless API entry point	Front door for Lambda endpoints	**Design Only**
IAM, CloudTrail, AWS Config	Account-level	Access control, audit, compliance	**Design Only**
Amazon ECR, ECS, EKS	Container track	Future roadmap	**Design Only**

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3. VPC Networking & Security Segmentation

3.1 Implemented network

The VPC uses the CIDR block 10.0.0.0/16 with DNS resolution enabled and default tenancy. Six subnets span two Availability Zones, each zone holding one public and two private subnets.

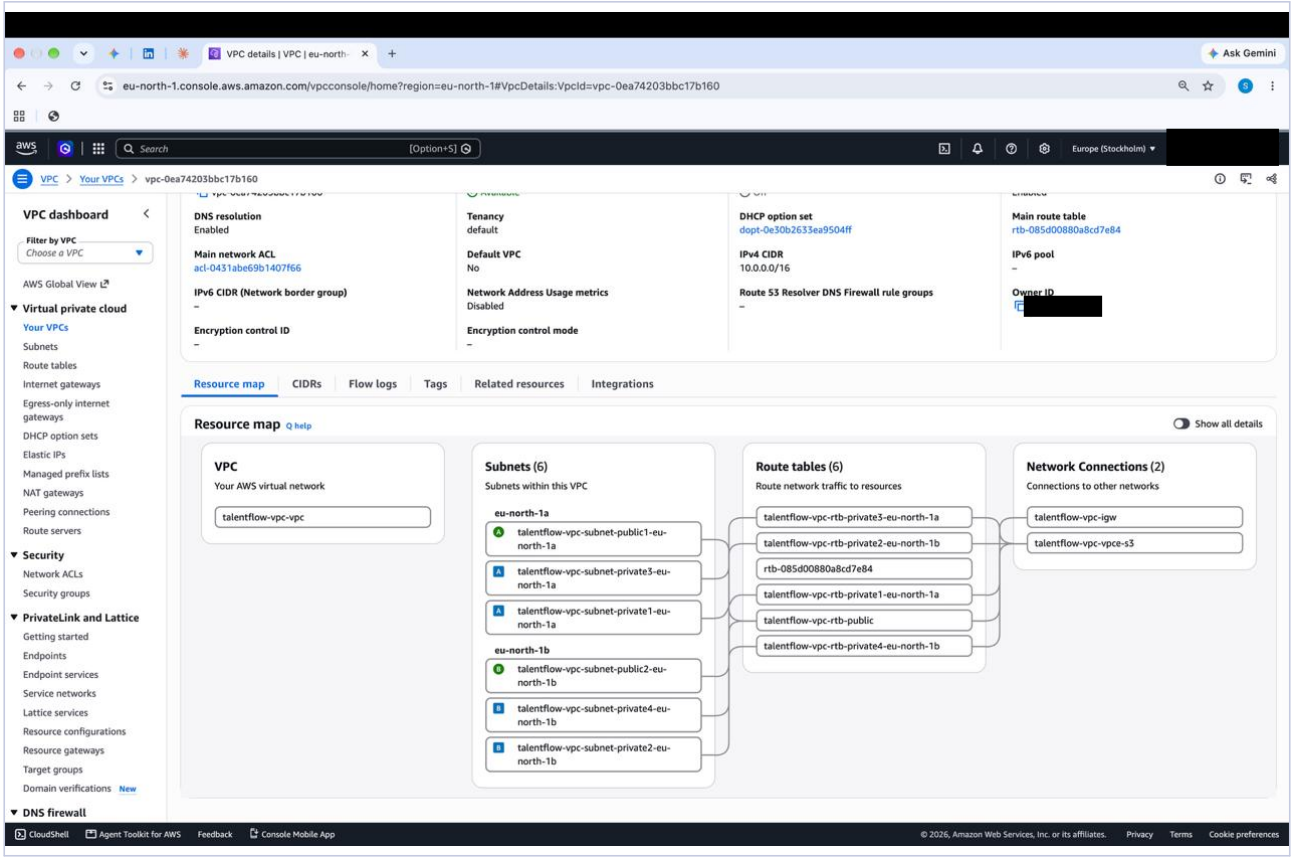
Availability Zone	Subnet	Tier	Route table
eu-north-1a	`talentflow-vpc-subnet-public1-eu-north-1a`	Public	`talentflow-vpc-rtb-public`
eu-north-1a	`talentflow-vpc-subnet-private1-eu-north-1a`	Private	`talentflow-vpc-rtb-private1-eu-north-1a`
eu-north-1a	`talentflow-vpc-subnet-private3-eu-north-1a`	Private	`talentflow-vpc-rtb-private3-eu-north-1a`
eu-north-1b	`talentflow-vpc-subnet-public2-eu-north-1b`	Public	`talentflow-vpc-rtb-public`
eu-north-1b	`talentflow-vpc-subnet-private2-eu-north-1b`	Private	`talentflow-vpc-rtb-private2-eu-north-1b`
eu-north-1b	`talentflow-vpc-subnet-private4-eu-north-1b`	Private	`talentflow-vpc-rtb-private4-eu-north-1b`

Six route tables are present, and two network connections are attached: the Internet Gateway and the S3 gateway endpoint.

Design decisions

- Public subnets hold internet-facing components; the database tier belongs in private subnets with no route to the internet.
- No NAT Gateway is deployed, which avoids its hourly and data-processing charges. The S3 gateway endpoint keeps S3 traffic private without NAT.
- Per-subnet private route tables allow egress control to be tightened later without redesign.

Figure 2: VPC resource map



VPC `talentflow-vpc-vpc` showing six subnets across eu-north-1a and eu-north-1b, six route tables, the Internet Gateway and the S3 gateway endpoint.

3.2 Security Group specification (Design Only)

The following least-privilege matrix is the specification for the compute and database tiers. It is not yet deployed.

Security Group	Direction	Protocol / Port	Source / Destination	Purpose
`talentflow-alb-sg`	Inbound	TCP 80, 443	0.0.0.0/0	Public HTTP and HTTPS ingress
`talentflow-alb-sg`	Outbound	TCP 80 / 3000	`talentflow-app-sg`	Forward traffic to backend instances
`talentflow-app-sg`	Inbound	TCP 80 / 3000	`talentflow-alb-sg`	Web traffic from the load balancer only
`talentflow-app-sg`	Inbound	TCP 22	Administrator IP (/32)	Restricted SSH administration
`talentflow-app-sg`	Outbound	TCP 5432	`talentflow-rds-sg`	PostgreSQL queries
`talentflow-app-sg`	Outbound	TCP 443	S3 gateway endpoint / AWS APIs	AWS API calls and resume ingestion
`talentflow-rds-sg`	Inbound	TCP 5432	`talentflow-app-sg`	Database queries from app instances only
`talentflow-rds-sg`	Outbound	None	None	No outbound internet access initiated

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4. Compute, High Availability & Auto Scaling (Design Only)

The compute tier is a production architecture specification. No EC2 instances, load balancer or Auto Scaling Group were left running, to avoid compute charges during this implementation phase. The Terraform

configuration declares the sizing variables (`instance_type`, `asg_min_size`, `asg_desired_capacity`, `asg_max_size`), but the compute resources themselves are not yet codified.

Specified configuration

Setting	Specification
Instance type	t3.micro, Amazon Linux 2023 (Free Tier eligible)
Deployment method	Launch template with User Data to install and start the application
Auto Scaling Group	Minimum 2, desired 2, maximum 4
Placement	Public subnets in eu-north-1a and eu-north-1b
Entry point	Application Load Balancer in the public subnets; listeners on HTTP 80 and HTTPS 443
Health checks	Target group probes a dedicated health endpoint; unhealthy instances are removed and replaced
Security groups	`talentflow-alb-sg` and `talentflow-app-sg` as defined in Section 3.2
Scaling	Target-tracking policy on average CPU utilisation

Placing instances in public subnets avoids NAT Gateway cost; the application security group accepts traffic only from the load balancer, which limits exposure. Moving the instances to private subnets with a NAT Gateway or VPC endpoints is the recommended hardening step for production. With two instances in two Availability Zones, the loss of one zone leaves the other serving traffic while the Auto Scaling Group restores capacity.

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5. Data Persistence & Storage Layer

5.1 Amazon RDS PostgreSQL (Design Only)

The managed database is a production architecture specification and is not deployed.

Setting	Specification
Engine and class	Amazon RDS for PostgreSQL, db.t3.micro
Network	Private DB subnet group across two Availability Zones
Public accessibility	No
Port and access	TCP 5432, reachable only from `talentflow-app-sg`
Data model	Users, candidates, jobs, applications, interviews, evaluations, notifications
Protection	Storage encryption and automated backups
Credentials	Held in a managed secret store, never in source code

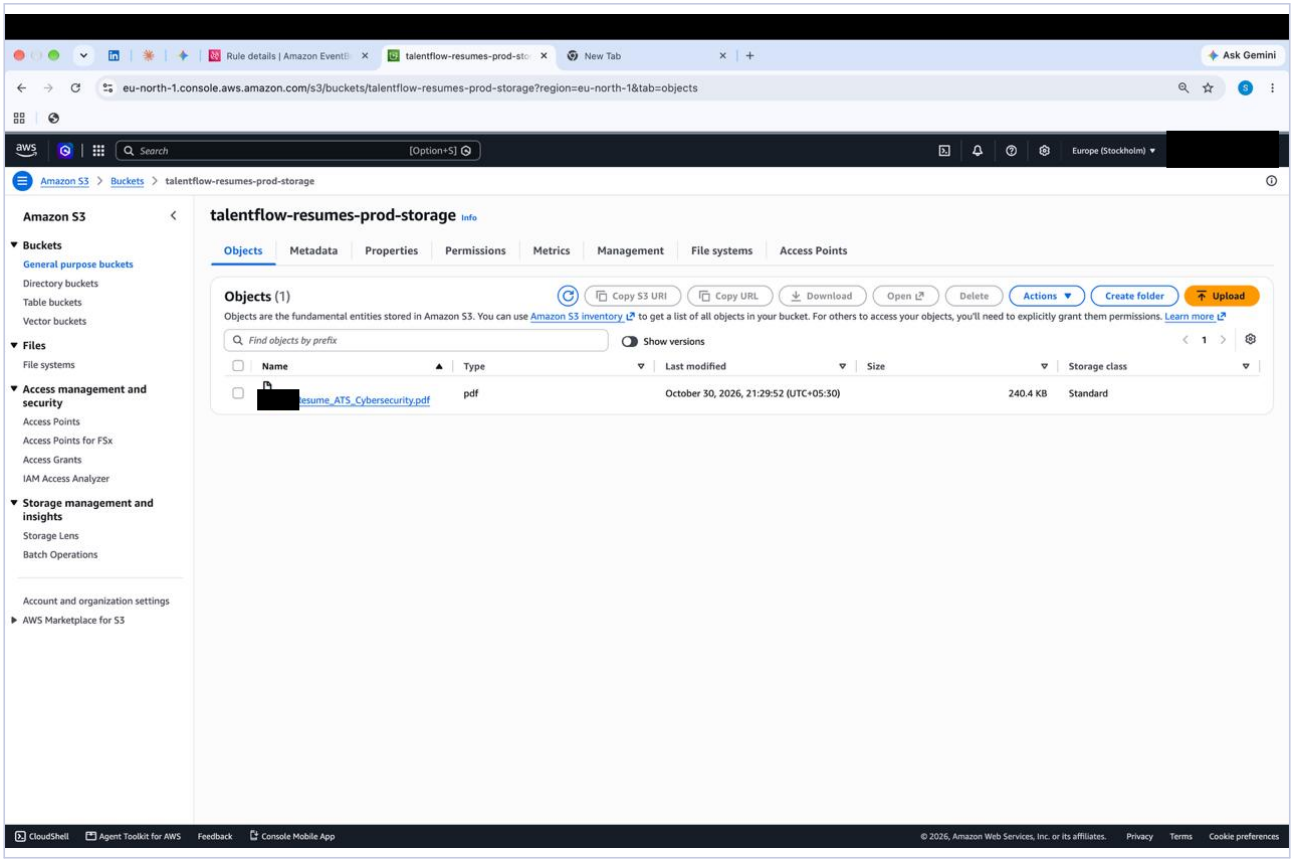
5.2 Amazon S3 (Implemented)

The bucket `talentflow-resumes-prod-storage` is deployed in eu-north-1. A resume PDF (240.4 KB, Standard storage class) was uploaded on 30 October 2026, confirming the storage path.

Control	Configuration
Block Public Access	All four settings enabled (defined in the Terraform configuration)
Encryption	SSE-S3 (AES256) default encryption (defined in the Terraform configuration)
Access	Private bucket; S3 gateway endpoint available inside the VPC

Object versioning and lifecycle rules are recommended hardening steps and are not part of the current configuration. Resumes contain personal data, so access should be restricted and retention should follow the organization's data-retention policy.

Figure 3: S3 bucket and resume upload



Bucket `talentflow-resumes-prod-storage` *containing the uploaded resume object.*

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6. Asynchronous & Event-Driven Processing

6.1 Processing flow

1. The application stores the resume in S3 and publishes a RESUME_UPLOADED event with source `talentflow.application`.
2. Rule `talentflow-resume-upload-rule` on the default event bus matches the event.
3. The rule delivers the event to the SQS queue `talentflow-resume-queue`.
4. Lambda function `talentflow-resume-processor` is triggered by the queue to process the resume.
5. Queue depth is monitored by CloudWatch and alerts reach recruiters through SNS.

The web request returns as soon as the event is published, so candidates never wait for background processing.

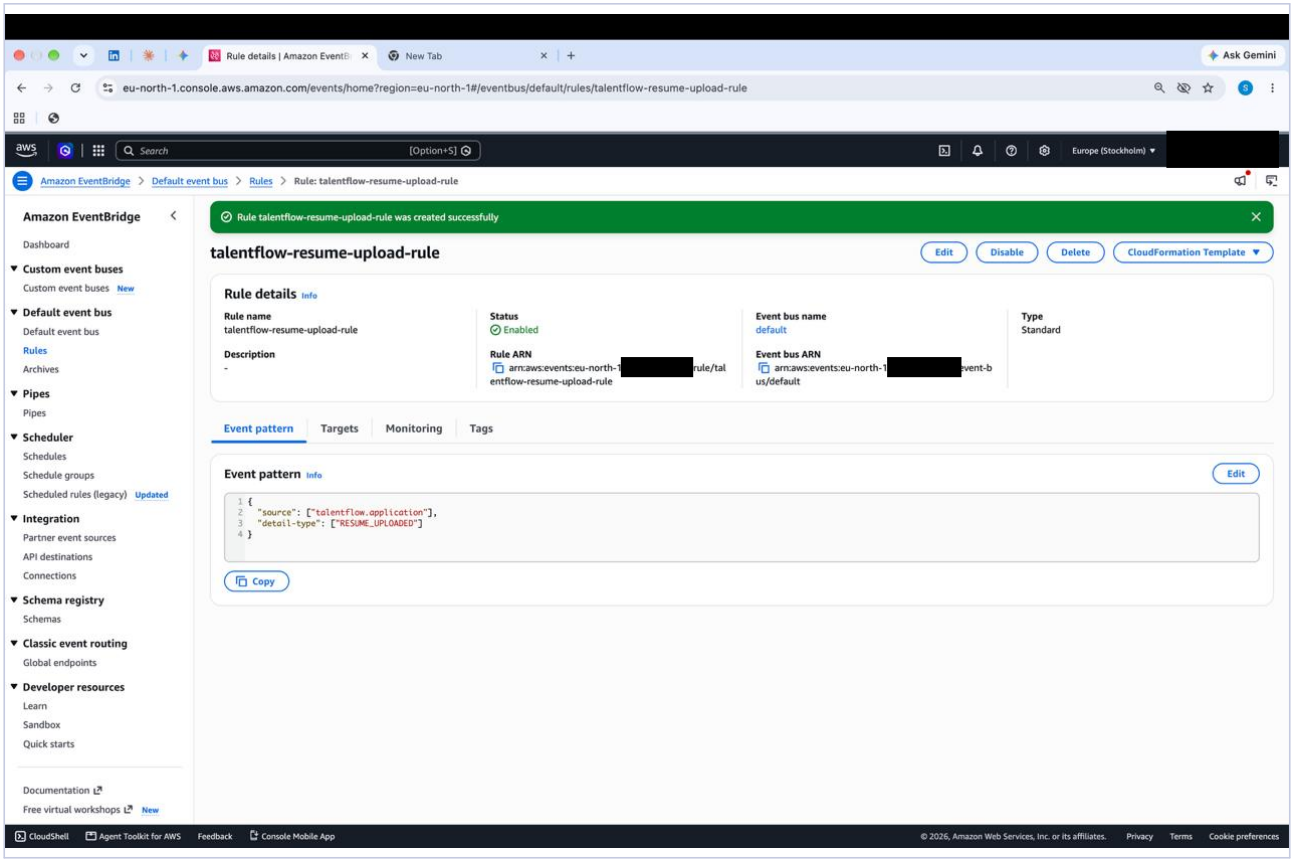
6.2 EventBridge rule

The rule is enabled on the default event bus (type Standard) with this event pattern:

```
{
  "source": ["talentflow.application"],
  "detail-type": ["RESUME_UPLOADED"]
}
```

Other business events (APPLICATION_CREATED, SCREENING_COMPLETED, CANDIDATE_SHORTLISTED, INTERVIEW_SCHEDULED, CANDIDATE_SELECTED) can each receive their own rule and target without changing the producer.

Figure 4: EventBridge rule configuration



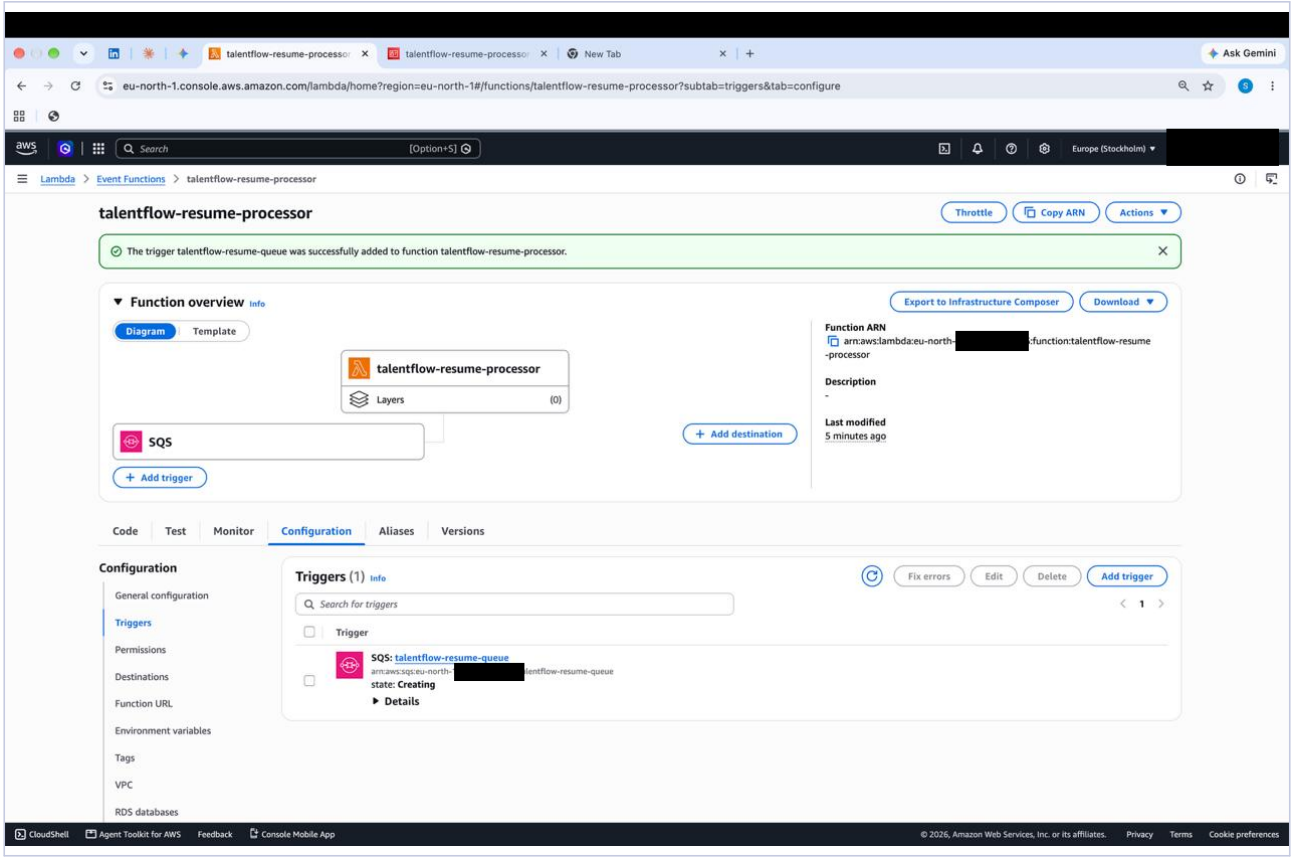
Rule `talentflow-resume-upload-rule` *created successfully and enabled, with the RESUME_UPLOADED event pattern.*

6.3 SQS queue and Lambda trigger

Setting	Configuration
Queue	`talentflow-resume-queue`, message retention 86,400 seconds (1 day), visibility timeout 300 seconds
EventBridge target	Rule target `SendToSQS` delivers matching events to the queue
Queue policy	Allows `sqs:SendMessage` only from the EventBridge service and only for the upload rule's ARN
Lambda trigger	SQS trigger from `talentflow-resume-queue` on `talentflow-resume-processor`

The queue policy follows least privilege: only the specific rule can send messages. The console confirmed the Lambda trigger was added successfully; the capture below shows it in its initial Creating state. The Lambda function and its trigger were configured in the console and are not yet part of the Terraform files. Execution of the function against a live message is not evidenced in this report.

Figure 5: SQS to Lambda trigger



Function `talentflow-resume-processor` *with the SQS trigger* `talentflow-resume-queue`.

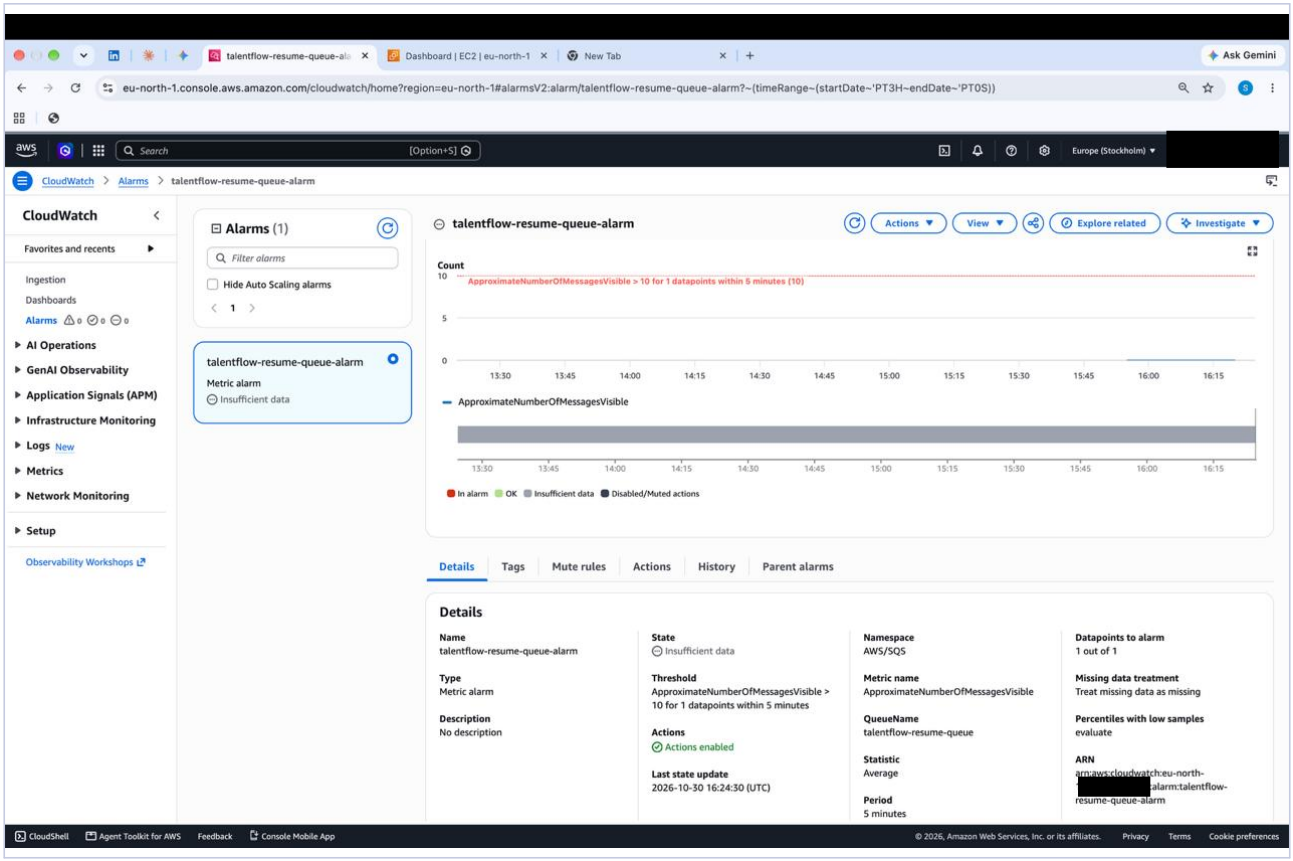
7. Observability, Monitoring & Alerts

7.1 Queue-depth alarm

Attribute	Value
Alarm name	`talentflow-resume-queue-alarm`
Metric	ApproximateNumberOfMessagesVisible (namespace AWS/SQS, queue `talentflow-resume-queue`)
Statistic and period	Average over 5 minutes
Condition	Greater than 10 for 1 out of 1 datapoints
Missing data	Treated as missing
Actions	Enabled; alarm action publishes to `talentflow-recruiter-notifications`

The alarm fires when more than ten resumes are waiting, indicating that processing is falling behind ingestion. The capture below was taken immediately after creation, when no datapoints had been recorded yet (state: Insufficient data).

Figure 6: CloudWatch alarm details

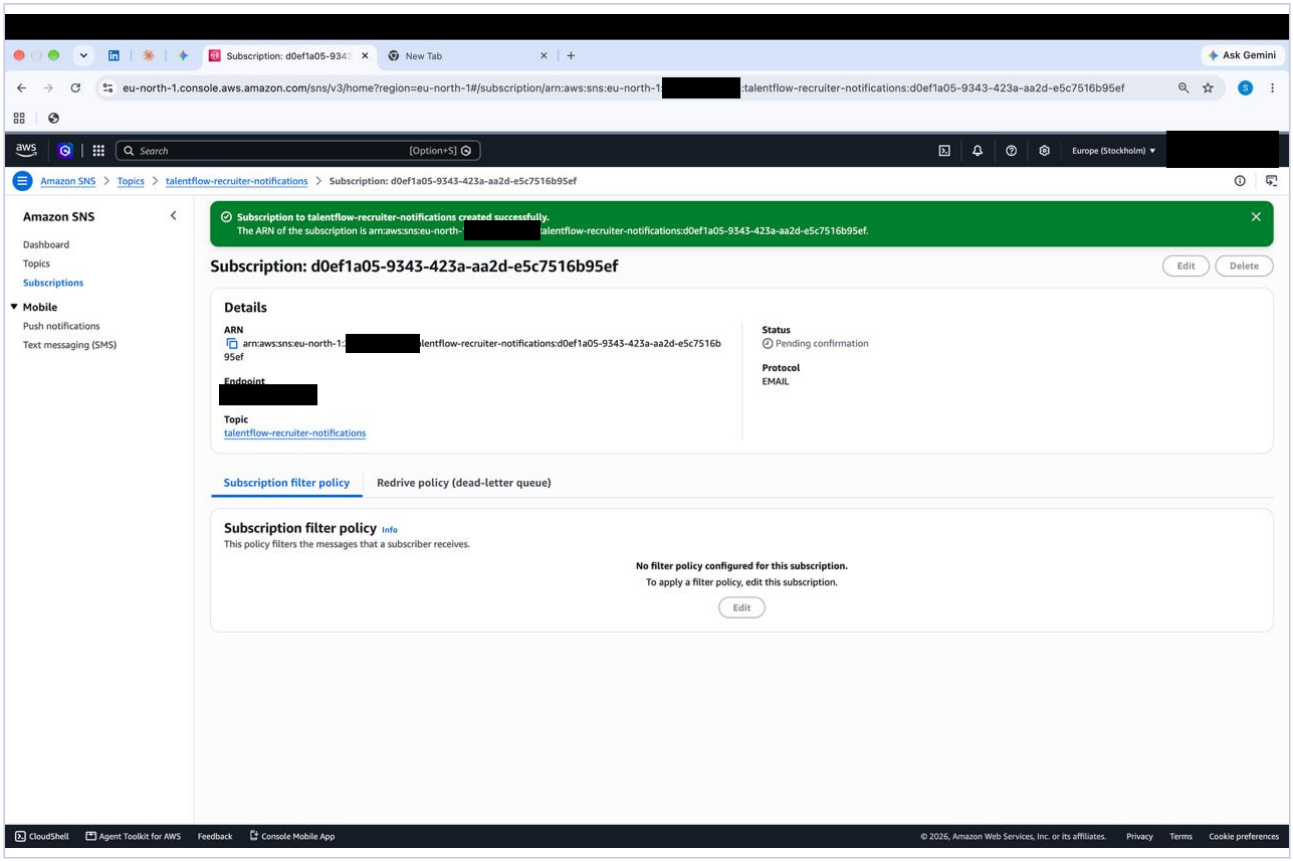


Alarm `talentflow-resume-queue-alarm` *with its threshold line at 10 on* `ApproximateNumberOfMessagesVisible`*.*

7.2 Recruiter notification path

Alarm action, SNS topic `talentflow-recruiter-notifications`, then an e-mail subscription (protocol EMAIL). The subscription was created successfully and showed Pending confirmation at capture; the recipient must confirm it from the AWS e-mail before messages are delivered. Message delivery is not evidenced in this report.

Figure 7: SNS subscription



Subscription to topic `talentflow-recruiter-notifications` *created successfully (status Pending confirmation at capture).*

7.3 Wider monitoring (Design Only)

When the compute and database tiers are deployed, CloudWatch will also cover EC2 and Auto Scaling metrics and application logs (through the CloudWatch agent), ALB request and error metrics, RDS metrics, and Lambda metrics and logs.

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8. Infrastructure as Code (Terraform)

8.1 Configuration layout

File	Responsibility
`main.tf`	Terraform version constraint, hashicorp/aws provider (~> 5.0), region and default tags
`variables.tf`	Region, VPC CIDR, instance type, Auto Scaling sizes, alarm threshold, notification e-mail
`vpc_and_compute.tf`	VPC, Internet Gateway, public and private subnets, S3 gateway endpoint, public route table and associations, S3 bucket with encryption and public-access block
`serverless_and_events.tf`	SQS queue and policy, EventBridge rule and target, SNS topic and subscription, CloudWatch alarm
`outputs.tf`	`vpc_id`, `s3_bucket_arn`, `sqs_queue_url`, `sns_topic_arn`, `cloudwatch_alarm_id`

The configuration defines 20 resources and applies default tags (Project, Environment, ManagedBy) to every taggable resource. Despite its file name, `vpc_and_compute.tf` currently codifies networking and storage only; EC2, ALB, Auto Scaling and RDS are not yet defined.

Scope note. The Terraform baseline codifies two public and two private subnets and one public route table. The console VPC in Figure 2 contains four private subnets and six route tables, so the console network extends beyond the codified baseline. Aligning the code with the console network, and adding the Lambda function and

its trigger, are listed as next steps in Section 15.

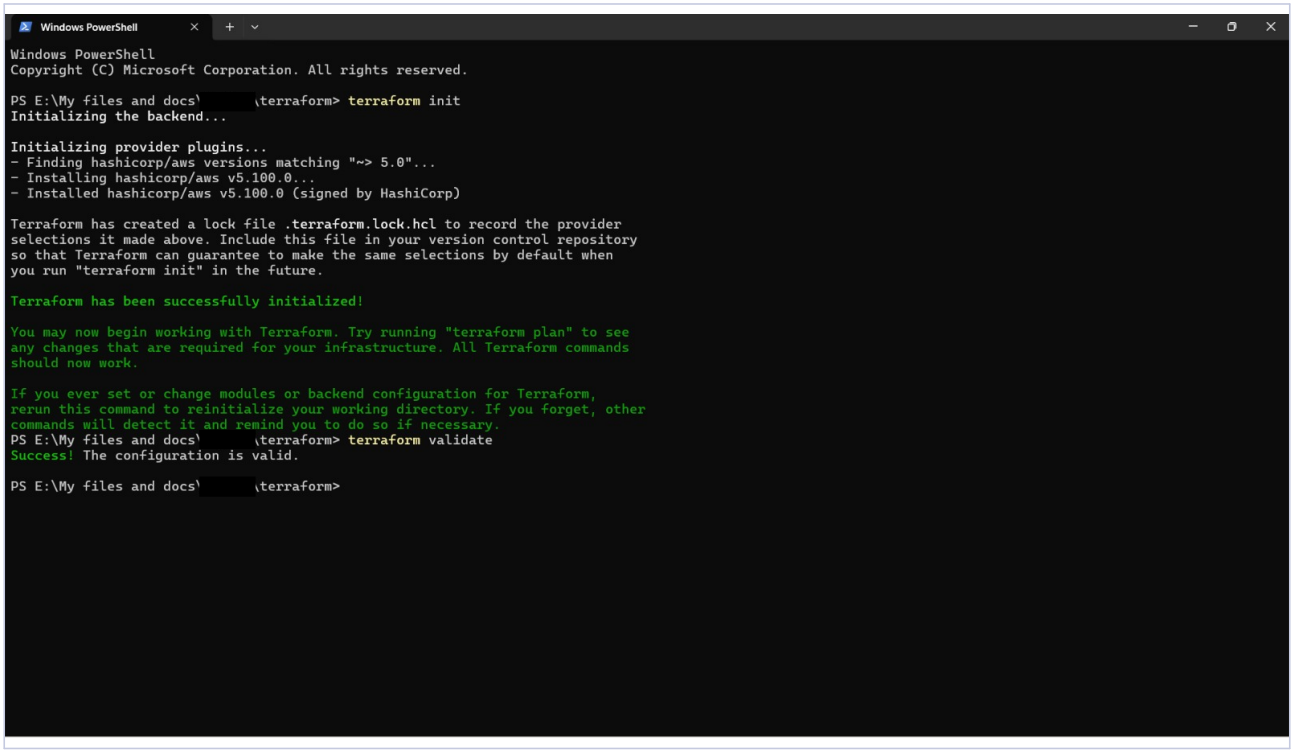
8.2 Verification

The terminal capture in Figure 8 shows:

- `terraform init` found hashicorp/aws matching ~> 5.0, installed v5.100.0 (signed by HashiCorp), created the dependency lock file `.terraform.lock.hcl` and reported successful initialisation.
- `terraform validate` returned: Success! The configuration is valid.

Initialisation and validation confirm the configuration's syntax, provider selection and internal consistency. Plan and apply output is not included in this report.

Figure 8: Terraform init and validate



PowerShell output showing successful initialisation and configuration validation.

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9. Security, Governance & Cost Optimization

9.1 Security controls

Domain	Control	Status
Storage	S3 Block Public Access and SSE-S3 encryption	Defined in validated Terraform
Messaging	SQS queue policy restricts senders to the specific EventBridge rule	Defined in validated Terraform
Network	Public and private subnet segmentation; S3 traffic over the gateway endpoint	**PASS** (Console Verified)
Network (compute and database)	Security group matrix in Section 3.2; private, non-public RDS	**Design Only**
Identity	Root account protected with MFA and not used daily; least-privilege roles for EC2, Lambda and Terraform	**Design Only**
Audit and compliance	CloudTrail API auditing and AWS Config rules	**Design Only**
Secrets	No credentials in source code; managed secret storage for database credentials	**Design Only**

9.2 Cost optimization

Lever	Approach
Phased deployment	Compute, database and edge tiers are specified but not left running, avoiding EC2, ALB, RDS and CloudFront charges during implementation
Compute sizing	t3.micro (Free Tier eligible) with a capped Auto Scaling maximum of four
Network	No NAT Gateway; S3 gateway endpoint avoids data-processing charges
Serverless	SQS, EventBridge, SNS and Lambda are pay-per-use and cost near zero when idle
Governance (**Design Only**)	AWS Budgets alerts, Cost Explorer reviews and Trusted Advisor checks
Cleanup	Resources removed with `terraform destroy` after testing

AWS resources are not assumed to be permanently free: Free Tier allowances are limited by time and usage, so unused resources should be removed promptly.

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10. Final Acceptance Criteria

#	Acceptance Criterion	Service(s)	Status	Basis
1	User accesses application	Browser, ALB, CloudFront	**Design Only**	Edge and compute tiers specified, not deployed
2	DNS resolves	Route 53	**Design Only**	Specified, not deployed
3	Traffic reaches AWS application layer	CloudFront	**Design Only**	Specified, not deployed
4	ALB distributes traffic	Application Load Balancer	**Design Only**	Specified, not deployed
5	Application runs successfully	EC2 and Auto Scaling	**Design Only**	Specified (t3.micro, 2 / 2 / 4), not deployed
6	Application communicates with RDS	Amazon RDS	**Design Only**	Specified (PostgreSQL, private), not deployed
7	Resume stored in S3	Amazon S3	**PASS** (Console Verified)	Figure 3: resume object in `talentflow-resumes-prod-storage`
8	Background processing uses event/queue architecture	Amazon EventBridge	**PASS** (Console Verified)	Figure 4: enabled rule matching RESUME_UPLOADED
9	Message queued	Amazon SQS	**PASS** (Console Verified)	Queue configured as the rule target and Lambda trigger source (Figures 4 and 5); message flow not captured
10	Lambda processes an event	AWS Lambda	**Design Only**	Function and SQS trigger exist (Figure 5); runtime processing not evidenced
11	Notification is generated	Amazon SNS	**PASS** (Console Verified)	Figure 7: topic and e-mail subscription created; delivery not evidenced
12	CloudWatch shows system health	Amazon CloudWatch	**PASS** (Console Verified)	Figure 6: alarm configured on queue depth with actions enabled
13	Security configuration	S3, SQS, IAM, CloudTrail, Config	**PASS** (Terraform Validated)	S3 public-access block, SSE-S3 and scoped SQS policy defined in validated code; IAM, CloudTrail and Config are **Design Only**
14	Infrastructure reproducible with Terraform	Terraform	**PASS** (Terraform Validated)	Figure 8: successful init and validate
15	Cost and resource usage explained	Budgets, Cost Explorer	**Design Only**	Strategy documented in Section 9.2; no billing data

Summary: 5 criteria are Console Verified (7, 8, 9, 11, 12), 2 are Terraform Validated (13, 14), and 8 are **Design Only** (1–6, 10, 15). The **Design Only** items are production architecture specifications retained for a later

deployment phase.

11. Project Deliverables Coverage

Deliverable area	Covered in	Status
VPC architecture, subnets, routing	Sections 2, 3.1	**PASS** (Console Verified)
Security Groups	Section 3.2	**Design Only**
High availability architecture	Section 4	**Design Only**
EC2, Application Load Balancer, Auto Scaling	Section 4	**Design Only**
Amazon S3	Section 5.2	**PASS** (Console Verified)
Amazon RDS	Section 5.1	**Design Only**
EventBridge, SQS, SNS	Sections 6, 7	**PASS** (Console Verified)
Lambda	Section 6.3	**PASS** (Console Verified): function and trigger; runtime **Design Only**
API Gateway	Section 2	**Design Only**
CloudWatch alarm	Section 7.1	**PASS** (Console Verified)
CloudWatch metrics and logs for compute and database	Section 7.3	**Design Only**
Route 53, CloudFront	Section 2	**Design Only**
Terraform configuration, variables, outputs	Section 8	**PASS** (Terraform Validated)
Docker, ECR, ECS/EKS	Section 14	**Design Only** (roadmap)
IAM design, least privilege	Section 9.1	**Design Only**
CloudTrail, security review	Section 9.1	**Design Only**
Budget configuration, cost analysis	Section 9.2	**Design Only**
Architecture diagram	Figure 1	Delivered
AWS service mapping	Section 2	Delivered
Deployment, security, cost and troubleshooting documentation	Sections 9, 12, 13	Delivered
Final README	`README.md` (separate file)	Delivered

12. Troubleshooting Guide

Symptom	Likely cause	Resolution
Alarm shows Insufficient data	No datapoints yet, or wrong queue dimension	Send test messages and confirm the QueueName dimension matches the queue
No e-mail from SNS	Subscription not yet confirmed	Click the confirmation link in the AWS message and check the spam folder
Lambda trigger stuck on Creating or Failed	Missing SQS permissions on the execution role	Add receive, delete and attribute permissions, then refresh
Messages processed repeatedly	Visibility timeout shorter than function duration	Keep the visibility timeout (300 s) above the Lambda timeout
EventBridge rule never triggers	Producer source or detail-type does not match the pattern	Match talentflow.application and RESUME_UPLOADED exactly
Events matched but queue stays empty	Queue policy does not allow the rule	Check the policy allows `sqs:SendMessage` from the rule's ARN
`terraform validate` passes but `apply` fails	Quota, permission or naming conflict, such as an existing bucket with the same name	Review the plan output and the failing resource's error message

13. Deployment Procedure

1. **Prepare:** install Terraform (1.5 or later) and the AWS CLI; sign in with a non-root IAM user or SSO role and confirm the account with `aws sts get-caller-identity`.
2. **Initialise:** run `terraform init` in the configuration folder (Figure 8).
3. **Validate:** run `terraform validate`; expect "Success! The configuration is valid."
4. **Plan:** run `terraform plan -out=tfplan` and review every resource to be created. Resources already created by hand in the console, such as the same-named S3 bucket, must be imported or removed first to avoid name conflicts.
5. **Apply:** run `terraform apply tfplan`, then `terraform output` to read the VPC ID, bucket ARN, queue URL, topic ARN and alarm ID.
6. **Post-deployment:** confirm the SNS e-mail subscription, publish a test RESUME_UPLOADED event, and confirm the alarm state in CloudWatch.
7. **Teardown:** run `terraform destroy` and check that no orphaned resources remain.

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14. Container Track (Roadmap)

Docker, Amazon ECR and ECS/EKS are an evaluation and future-roadmap track. No container image, ECR repository or ECS/EKS deployment was created; the application is specified to run on EC2 behind an Application Load Balancer using a launch template and User Data. The migration path is: build a Docker image, push it to a private ECR repository, and run it on ECS (Fargate) behind the existing load balancer, or on EKS where Kubernetes features are required, keeping the same VPC, database, storage and event pipeline.

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15. Limitations & Next Steps

Item	Current state	Next step
Compute tier	Specified only	Codify and deploy launch template, ALB and Auto Scaling Group
Database	Specified only	Codify and deploy RDS PostgreSQL in the private subnet group
Edge and API	Specified only	Add Route 53, CloudFront and API Gateway
Lambda	Created in the console; not in Terraform	Import or define the function, role and event source mapping in code
Network	Console VPC has four private subnets and six route tables; Terraform defines two private subnets and one route table	Align the code with the console network
Pipeline testing	Configuration verified; live message flow and notification delivery not captured	Publish a test event, capture the Lambda log and the delivered e-mail
Governance	Specified only	Enable CloudTrail, AWS Config and a Budget, and capture the results
S3 hardening	Versioning and lifecycle not configured	Add versioning and lifecycle rules

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16. Conclusion

TalentFlow AI now has a verified AWS foundation: a segmented VPC, an encrypted private resume bucket, an event-driven ingestion pipeline built from EventBridge, SQS and a Lambda trigger, recruiter alerting through CloudWatch and SNS, and a Terraform configuration that initialises and validates successfully. The compute tier, managed database and edge delivery layer are fully specified as the production architecture and were

deliberately left undeployed to avoid ongoing charges. Section 15 lists the steps needed to move those specifications into running infrastructure.